Engineering

Metric bolt shear strength for a layman

Asked 8 months ago Modified 8 months ago Viewed 8k times

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As a layman, how can I know the shear strength of a particular size bolt?

For example, if I want to attach something to the internal wall of my van (1mm stainless steel) using a rivnut and standard metric bolt and it should hold up to 50 kg, how would I figure out the right size and grade to use? Nothing specialty, just standard stuff a layman could get at a typical hardware store.

In my specific case I'm installing wooden cabinets on the internal walls of my Fiat Ducato van, but I'd like as general an answer as reasonable.

shear

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asked Sep 12, 2021 at 16:22



Robert S. Barnes
163 1 5

9 The shear strength of the bolt is likely not going to be the limiting factor. Pullout force of the rivnut or flex in the sheet metal are more relevant. Just use plenty of whatever size is most convenient to spread the load. – Jonathan R Swift Sep 12, 2021 at 18:07

1 Honestly, having fitted out a van myself, the limiting factor is likely to be vibration and loosening. Something, somewhere, will work loose, perhaps in the construction of the cabinets themselves. Differential thermal expansion between wood and steel isn't your friend either – Chris H Sep 13, 2021 at 13:36

4 Anybody can design a bridge that stands. But it takes an engineer to design a bridge that *just barely* stands. **Don't be an engineer. Overdo it.** – dotancohen Sep 13, 2021 at 15:46

1 @dotancohen You'd have to be an aerospace engineer for that to hold. Other engineers use *safety factors* that are noticeably higher than 1. – TooTea Sep 14, 2021 at 9:10

1 @TooTea sure, but the point is they don't use a safety factor of 50, because that would be prohibitively expensive and of marginal real safety benefit, but spending $50 instead of $5 on your van is less so – llama Sep 14, 2021 at 15:48

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12 is the friction of the two surfaces that support the load. This is the principle that holds wheels

on automobile hubs. The friction of hub to the wheel takes the forces, not the shear strength of the studs.

For holding a cabinet up, just use appropriately sized bolts and you will be fine.

ETA: As @r13 comments, the engineering work of using friction of the joined pieces is to determine the required friction, the required bolt tension, number of bolts, the required bolt torque to get that tension, bolt specifications, etc. I consider this beyond the layman level.

Share Improve this answer Follow edited Sep 13, 2021 at 16:56 answered Sep 13, 2021 at 0:43



Tiger Guy
4,417 4 15

1 The friction type bolting is preferable over the bearing type as it provides additional safety factors against mishaps. However, the proper torque and resulting strength level usually remain a mystery for most non-technical people. – r13 Sep 13, 2021 at 1:30

2 If your bolts are rated for the shear load, you don't have to worry about slip-critical joints and their complexity. Maybe you've got enough tension to keep the joint from slipping, maybe you don't. It's a lot easier to add up a dozen shear ratings off the spec sheet (then give yourself a safety margin since different bolts might take different loads) than to worry about friction levels. Since the correct answer shouldn't be "go slip-critical", you should probably explain what an appropriately-sized bolt is. – MichaelS Sep 13, 2021 at 21:09

14

For complete layman's terms, an m6 bolt can hold up an elephant. For 10 kg, the bolts won't be the weak link, it will be whatever you are bolting into. Use 3 or 4 and you will probably be fine

The fine print: I'm considering an average female Asian elephant (6000 lb or 2700 kg), grade 8 bolts (the expensive "top shelf" bolts) , double shear, and no additional safety factor.

Share Improve this answer Follow edited Sep 13, 2021 at 10:31 answered Sep 12, 2021 at 22:04



Daniel K
2,511 5 13

Talk about the elephant in the room... – manassehkatz-Moving 2 Codidact Sep 15, 2021 at 15:54

8

The first thing to know is the "grade" of the bolt is available to you. Then use the appropriate table (similar to the one below) from the standard code to determine the loading capacity of the bolt. Note that your application shouldn't involve "bending" and "twisting", for which the

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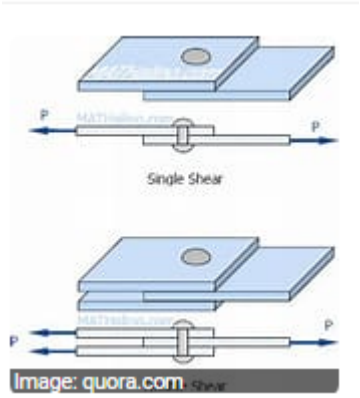
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Bolt Loading Capacities. Non-preloaded Ordinary Bolts. Grade 8.8 Bolts per. S355

Diameter of Bolt	Tensile Stress Area A_t	Tension Capacity		Shear Capacity		Bearing Capacity in kN (Minimum of P_{bb} and P_{bs}) End distance equal to 2 x bolt diameter Thickness in mm of material passed through									
		Nominal 0.8 A_t p_t	Exact A_t p_t	Single Shear	Double Shear										
		P_{nom}	P_t	P_s	2 P_s										
mm	mm^2	kN	kN	kN	kN	5	6	7	8	9	10	12	15	20	25	30
12	84.3	37.8	47.2	31.6	63.2	33.0	39.6	46.2	52.8	59.4	66.0	79.2	99.0	132	165	198
16	157	70.3	87.9	58.9	118	**44.0**	**52.8**	61.6	70.4	79.2	88.0	106	132	176	220	264
20	245	110	137	91.9	184	**55.0**	**66.0**	**77.0**	**88.0**	99.0	110	132	165	220	275	330
22	303	136	170	114	227	**60.5**	**72.6**	**84.7**	**96.8**	**109**	121	145	182	242	303	363
24	353	158	198	132	265	**66.0**	**79.2**	**92.4**	**106**	**119**	**132**	158	198	264	330	396
27	459	206	257	172	344	**74.3**	**89.1**	**104**	**119**	**134**	**149**	178	223	297	371	446
30	561	251	314	210	421	**82.5**	**99.0**	**116**	**132**	**149**	**165**	**198**	248	330	413	495

Single Shear vs Double Shear

In general, double shear has twice the shear area compared to single shear.



Note: When you need the information on the bolt with a smaller size, either try to get the information from the appropriate code, or, most directly, read the instructions of the product, which usually shall indicate the safe working load.

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Allowable Screw Connection Capacity (lbs)											
#6 Screw			#8 Screw			#10 Screw			#12 Screw		
(Pss = 643 lbs, Pts = 419 lbs)			(Pss= 1278 lbs, Pts = 586 lbs)			(Pss= 1644 lbs, Pts = 1158 lbs)			(Pss= 2330 lbs, Pts = 2325 lbs)		
0.138" dia, 0.272" Head			0.164" dia, 0.272" Head			0.190" dia, 0.340" Head			0.216" dia, 0.340" Head		
Shear	Pull-Out	Pull-Over	Shear	Pull-Out	Pull-Over	Shear	Pull-Out	Pull-Over	Shear	Pull-Out	Pull-Over
44	24	84	48	29	84	52	33	105	55	38	105
82	37	127	89	43	127	96	50	159	102	57	159
95	40	140	103	48	140	111	55	175	118	63	175
151	61	140	164	72	195	177	84	265	188	95	265
214	79	140	244	94	195	263	109	345	280	124	345
214	100	140	344	118	195	370	137	386	394	156	433
214	125	140	426	149	195	523	173	386	557	196	545
214	140	140	426	195	195	548	246	386	777	280	775
214	140	140	426	195	195	548	301	386	777	342	775
214	140	140	426	171	195	534	198	386	569	225	625
214	140	140	426	195	195	548	249	386	777	284	775
214	140	140	426	195	195	548	356	386	777	405	775
214	140	140	426	195	195	548	386	386	777	494	775

Conversion: 1 kgf = 2.2 lbs

Gauge #, mm:

#6 - 3.5mm, #8 - 4mm, #10 - 5mm, #12 - 5.5mm

Share Improve this answer Follow edited Sep 12, 2021 at 22:16

answered Sep 12, 2021 at 19:03



r13

7,293 3 6 25

1 Good answer, although perhaps a table showing M4 thru M12 would be more appropriate for the task at hand! – Jonathan R Swift Sep 12, 2021 at 19:56

@JonathanRSwift This answer is meant to provide an example of how and where to get one's hand on the loading capacity of a fastener of a specific grade. Unfortunately, the data source only made the information of the structural grade bolts available. For smaller sizes (machine bolts), the OP needs to find the information through other venues. – r13 Sep 12, 2021 at 21:20

You can't find data on M4 to M8? – Solar Mike Sep 13, 2021 at 5:02

@SolarMike r13 presumably comes from either the USA or Liberia, judging by the choice of units in his/her posts. (But from his/her frequent unidiomatic use of "shall" I might have guessed Germany instead...) – alephzero Sep 14, 2021 at 19:50

@SolarMike No. Not for the structural bolt which the table covers. I guess a bolt smaller than M12 may not be permitted for structural use in the British, you can conveniently check it out. – r13 Sep 14, 2021 at 20:56

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where:

where:

αv is a coefficient that takes values αv = 0.6 for bolt classes 4.6, 5.6, 8.8 or αv = 0.5 for bolt classes 4.8, 5.8, 6.8 and 10.9. When the shear plane passes through the unthreaded part of the bolt αv = 0.6.

- fub is the ultimate tensile strength of the bolt depending on the bolt class (see table above)
- A is the appropriate area for shear resistance. When the shear plane passes through the threaded part of the bolt A is equal to the tensile stress area of the bolt As. When the shear plane passes through the unthreaded part of the bolt A is equal to the gross cross-sectional area of the bolt Ag.
- γM2 is the partial safety factor for the resistance of bolts in accordance with EN1993-1-8 §2.2(2) Table 2.1 and the National Annex. The recommended value in EN1993-1-8 is γM2 = 1.25.

For small DIY work i would recommend using 0.3 Ag multiplied by fu of the bolt you use. but you have to mind if bolt is too strong it can tear or warp the material.

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answered Sep 12, 2021 at 17:24



kamran
20.9k 2 16 34

2

I will start by highlighting the main points (some were also found in other answers). I will additionally provide a short calculation for the *tensile strength* (and *shear strength*) very quickly by knowing the **type** (e.g. M6) and **grade** (e.g 10.9). and below I will providing a layman's calculation for a metric bold.

In the numerical calculation I will use a very small, and very low grade bolt to show case that the bolts usually aren't the weak link in this case.

the important parts for a layman

- (tiger guy's): **Bolts shouldn't be used in shear** (they can be used but in most scenarios are not suggested)
- for 50 kg any metric bolt should be able to hold.
- (Daniel K) In many cases, bolts are not the weak link in the assembly.

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o.o.

- The first number multiplied by 100 is the UTS of the material MPa
- The second number (with the dot) is the ratio of the yield stress to the UTS of the material.

So for an 5.6 bolt the UTS is $5 \cdot 100 = 500MPa$, and the yield stress is $0.6 \cdot \sigma_{UTS} = 300MPa$. The followonig table has some of the most common bolt grades.

	UTS	yield stress
4.6	400 MPa	240 MPa]
5.6	500 MPa	300 MPa
8.8	800 Mpa	640 MPa
10.9	1000 MPa	800 MPa
etc	..	..

So the *tensile strength* up to yield strength of an M4 4.6 is:

$$A \cdot \sigma_y = \frac{\pi \cdot (4[mm])^2}{4} \cdot 240 \left[\frac{N}{mm^2}\right] = 3015N$$

Or approximately 300 kg.

Calculation for bolt shear strength

For the shear strength, you can approximate the shear yield stress to 0.5 of the tensile yield stress (this is a **very** conservative approximation if you compare it to tables from Eurocode). So the shear load should be in the order of (at least) 150 kg.

Please note that I used in the example one of the smaller metric bolts M4 (its hard to find a wrench them), at the lowest grade, and it still was capable of holding (At least) 3 times the 50 kg load.

Share Improve this answer Follow edited Sep 13, 2021 at 8:33

answered Sep 13, 2021 at 5:54



NMech
22.6k 2 29 66

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I have often just pried rivnuts out when the bolts have been seized.

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answered Sep 12, 2021 at 16:59



Solar Mike
13.6k 1 18 27

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